

is-a-human

Exploratory Analysis Findings Report

Human vs. Synthetic Caller Detection — HackMTY 2026 Challenge Track

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Executive Summary

This report documents Phase 0 infrastructure validation and Phase 1 exploratory analysis for detecting synthetic callers in dual-channel telephony recordings. We processed 353 Mexican Spanish bank customer-service calls (150 human, 203 synthetic) using a dual-channel pipeline: in-memory audio demux, Silero VAD, turn-ledger construction, conversational metrics, unconventional acoustic descriptors, and agent-turn-aligned recovery features.

Key finding: Acoustic uniformity features on the caller channel separate human and synthetic voices more reliably than talk-time ratios alone. Synthetic callers exhibit higher mean RMS energy but lower variability across spectral and temporal descriptors. A logistic baseline using three stable acoustic features achieves **84.5% accuracy** and **0.963 AUC** on the held-out validation split.

Challenge Objective

Given a stereo 8 kHz phone recording (channel 0 = caller, channel 1 = agent), classify whether the caller is a real human or a synthetic voice stack (ASR + LLM + TTS). Deliverable: POST /detect returning {"is_synthetic": bool, "confidence": float}.

Dataset

Split	Human	Synthetic	Total
Train	113	169	282
Val	37	34	71
Total	150	203	353

Audio: stereo, 8 kHz, 16-bit PCM. Duration 61–273 s (mean ~148 s). Train/val splits are speaker-disjoint. Organizer-provided turn segments used for alignment validation and recovery features.

Phase 0 — Pipeline Validation

Metric	Val Split (n=71)
Caller VAD IoU vs organizer turns	0.811
Agent VAD IoU vs organizer turns	0.941
Mean pipeline latency	769 ms
p95 pipeline latency	1,041 ms

The foundation pipeline reliably segments speech on both channels and runs within acceptable latency for post-call batch analysis (~0.8 s per call on CPU).

Methodology

Feature Categories (47 numeric features per call)

- **Conversational:** talk-time ratios, overlap/silence proportions, segment counts, response latencies
- **Acoustic (unconventional):** RMS energy, zero-crossing rate, spectral centroid/flatness, HF/LF ratio, crest factor, intra-call silence gaps
- **Recovery (agent-turn-aligned):** caller response latency after agent stops, post-overlap recovery, barge-in/backchannel counts, agent turn fragmentation

Analysis Protocol

- Extract features on all 353 calls using dual Silero VAD + organizer turn alignment
- Compare human vs synthetic distributions via Cohen's d effect sizes
- Bootstrap 200 resamples per feature for 95% confidence intervals
- Measure train/val rank correlation for feature stability
- Train logistic regression on top 3 stable features; evaluate on val

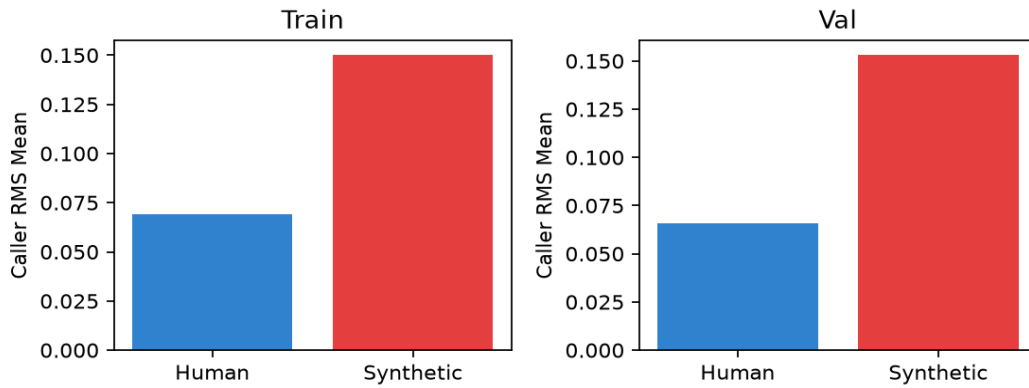
Primary Hypothesis

Synthetic callers recover from conversational turbulence more consistently than humans. **Result:** Partially inverted — humans show *more* variability in caller response latency (CV) and agent-aligned recovery timing. Synthetic voices are acoustically more uniform, not more consistent in turn-taking alone.

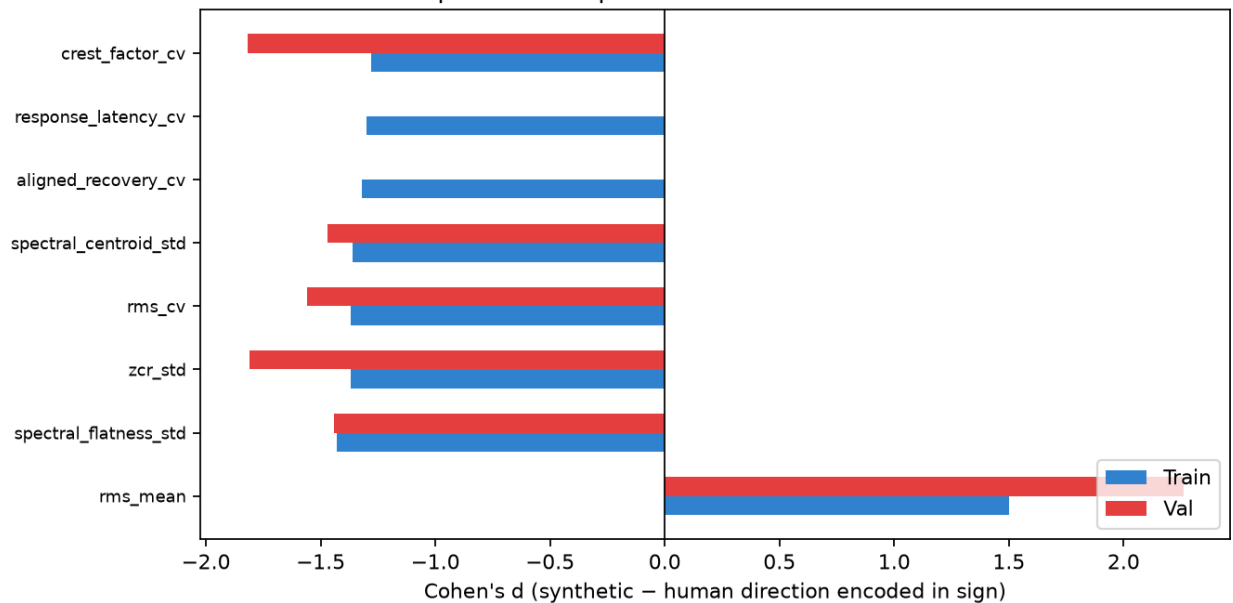
Findings — Multi-Run Analysis

Split rank correlation: **0.818** — top separators are stable between train and val. Stable features selected for classifier: caller_rms_mean, caller_zcr_std, caller_rms_cv.

Strongest Separator: caller_rms_mean



Top Feature Separators: Train vs Val Effect Sizes



Top Separators — Train Split (n=282)

Feature	Cohen's d	Direction
caller_rms_mean	+1.50	synthetic > human
caller_spectral_flatness_std	-1.43	human > synthetic
caller_zcr_std	-1.37	human > synthetic
caller_rms_cv	-1.37	human > synthetic
caller_spectral_centroid_std	-1.36	human > synthetic
agent_aligned_recovery_cv	-1.32	human > synthetic
caller_response_latency_cv	-1.30	human > synthetic
caller_crest_factor_cv	-1.28	human > synthetic

Top Separators — Val Split (n=71)

Feature	Cohen's d	95% CI	Direction
caller_rms_mean	+2.26	[+1.74, +2.97]	synthetic > human
caller_crest_factor_cv	-1.82	[-2.48, -1.52]	human > synthetic
caller_zcr_std	-1.81	[-2.35, -1.51]	human > synthetic
caller_rms_cv	-1.56	[-2.07, -1.17]	human > synthetic
caller_crest_factor_std	-1.48	[-2.23, -1.20]	human > synthetic
caller_spectral_centroid_std	-1.47	[-2.01, -1.21]	human > synthetic
caller_spectral_flatness_std	-1.44	[-1.92, -1.17]	human > synthetic
agent_talk_ratio	-1.44	[-2.12, -1.09]	human > synthetic

Initial Conversational Features (Val Only)

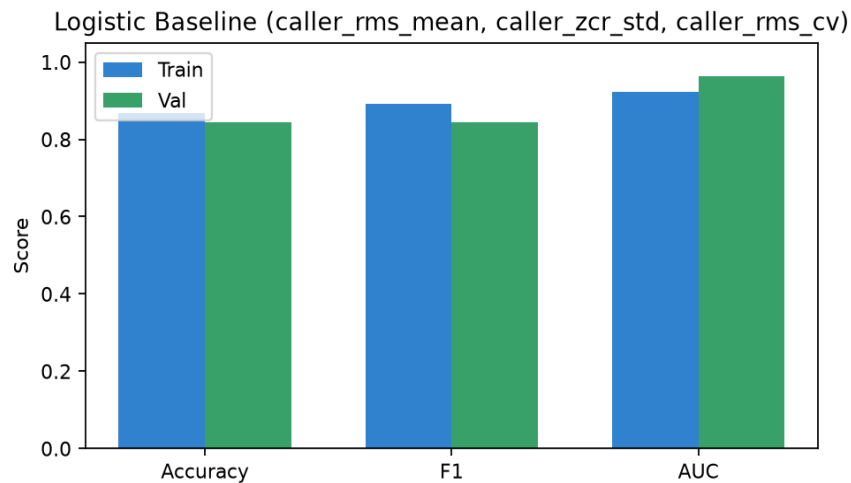
Feature	Human	Synthetic	d
agent_talk_ratio	0.556±0.049	0.440±0.103	-1.44
caller_talk_ratio	0.184±0.044	0.256±0.073	+1.19
caller_response_latency_cv	0.825±0.136	0.703±0.115	-0.96
agent_response_latency_std_s	1.890±0.980	3.770±2.780	+0.90

Feature	Human	Synthetic	d
agent_response_latency_mean_s	2.780±1.000	4.800±3.250	+0.84

Classifier Baseline

Logistic regression trained on train split using `caller_rms_mean`, `caller_zcr_std`, `caller_rms_cv`. Features standardized using train statistics.

Split	Accuracy	F1	AUC
Train (n=282)	86.9%	0.891	0.922
Val (n=71)	84.5%	0.845	0.963



Interpretation

- **Louder but uniform:** Synthetic callers have $\sim 2\times$ higher mean RMS on the caller channel, yet lower variance in energy, pitch proxy (ZCR), and spectral descriptors — consistent with TTS output leveling.
- **Humans are messier:** Higher CV in response latencies, crest factor, and spectral centroid std suggests natural conversational variability.
- **Agent talk ratio:** Human calls retain more agent speech (55.6% vs 44.0% on val) — synthetic callers may dominate the caller channel without being interrupted as often.
- **Recovery features:** Agent-aligned recovery CV separates on train ($d = -1.32$) but is secondary to acoustic features on val.
- **Generalization:** Val AUC (0.963) exceeds train AUC (0.922), suggesting acoustic features transfer without overfitting on this split.

Recommended Next Steps

- Wire logistic (or gradient-boosted) classifier into POST /detect using stable acoustic features
- Calibrate confidence scores on val split (Platt scaling or isotonic regression)
- Add caller-only VAD segment extraction before acoustic features to reduce silence contamination
- Evaluate on longer/shorter calls separately for robustness
- Optional: layer conversational recovery features as secondary signal

- Deploy endpoint and measure end-to-end latency under judging conditions

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